

RAJNEESH CHAUHAN

Data Scientist

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PROFESSIONAL SUMMARY

Production-focused Data Scientist skilled in building end-to-end ML pipelines, deploying containerized APIs, and designing Retrieval-Augmented Generation (RAG) systems using Python, SQL, XGBoost, and MLflow. Delivers measurable business impact — ROC-AUC of 0.87 on churn prediction, 87% precision on fraud detection, and production-grade RAG Q&A system with sub-100ms inference. Strong MLOps foundation with Docker, FastAPI, DVC, CI/CD, Prometheus, and AWS; proven ability to ship interpretable, production-grade models from EDA to deployment.

TECHNICAL SKILLS

Programming Languages: Python, SQL, MySQL

ML & DL Frameworks: Scikit-learn, XGBoost, LightGBM, TensorFlow, Keras, LSTM, Neural Networks, SHAP, Optuna, GridSearch

Data Science: Machine Learning, Deep Learning, EDA, Feature Engineering, Feature Selection, Time Series Forecasting, A/B Testing, Hypothesis Testing, Statistical Modeling, NLP, Text Classification

Big Data & Tools: Apache Spark, PySpark, Databricks, MLflow, DVC, Docker, FastAPI, Flask, GitHub Actions, CI/CD, AWS (EC2, S3), DagsHub, Prometheus, Grafana, Tableau, Power BI

Libraries & Databases: Pandas, NumPy, Matplotlib, Seaborn, TF-IDF, Word Embeddings, Git, GitHub, MySQL

PROFESSIONAL EXPERIENCE

Data Science Intern

Aug 2025 – Oct 2025

Skillied Mentor

Remote

- Engineered targeted feature pipelines and applied EDA on 50,000+ imbalanced records using Pandas, NumPy, and Seaborn; boosted model F1-score by 12% over baseline through rigorous feature selection and data preprocessing.
- Designed and deployed real-time prediction APIs using Flask and FastAPI with integrated model monitoring via Prometheus; reduced post-deployment false positive rate by 8% by proactively detecting and responding to prediction drift.

Technologies Used: Python, Pandas, NumPy, Flask, FastAPI, Prometheus, Matplotlib, Seaborn

PROJECTS

Customer Churn Prediction & Business Insights [\[GitHub\]](#) XGBoost, LightGBM, MLflow, DVC, FastAPI, Docker, SHAP, Optuna, Databricks,

DagsHub

Jan 2026

- Achieved ROC-AUC of 0.87 by designing a full ML data pipeline — training XGBoost, LightGBM, Random Forest, and Logistic Regression with Optuna-driven hyperparameter tuning and MLflow experiment tracking on DagsHub.
- Shipped a containerized FastAPI serving layer via Docker to DockerHub, backed by a DVC-managed data pipeline on DagsHub S3, achieving full reproducibility across all pipeline runs.
- Surfaced contract type, tenure, and monthly charges as top churn drivers through SHAP interpretability, translating model outputs into a revenue impact model for actionable retention strategies.
- Leveraged Databricks for scalable distributed data processing and collaborative ML experiment management, accelerating model iteration and team reproducibility across the pipeline.

Real-Time Fraud Detection API with Full MLOps Pipeline [\[GitHub\]](#) XGBoost, LightGBM, FastAPI, Docker, MLflow, DVC, Prometheus,

Grafana, CI/CD, GitHub Actions

Mar 2026

- Achieved 87% precision on highly imbalanced ULB Credit Card fraud dataset by training XGBoost and LightGBM with SMOTE oversampling, ensemble stacking, and Optuna-driven hyperparameter tuning tracked via MLflow.
- Engineered a real-time prediction API using FastAPI + Docker (pushed to DockerHub) with sub-50ms inference latency, exposing a /predict endpoint for live transaction scoring in production.
- Instrumented end-to-end model monitoring with Prometheus and Grafana — tracking prediction drift, false positive rate, and feature distribution shifts — triggering automated retraining via GitHub Actions CI/CD when drift exceeded threshold.
- Versioned datasets and model artifacts with DVC on AWS S3, ensuring full pipeline reproducibility and enabling rollback to any prior model version within minutes.

DocMind — AI-Powered PDF Q&A Chatbot using RAG [\[GitHub\]](#) Python, LangChain, HuggingFace, sentence-transformers, ChromaDB,

LLaMA 3.1 (Groq), Streamlit, RAG, NLP, Docker

Feb 2026

- Engineered a full Retrieval-Augmented Generation (RAG) pipeline — extracting and chunking PDF documents with PyMuPDF, embedding with HuggingFace all-MiniLM-L6-v2 sentence-transformers, indexing into ChromaDB for top-k cosine similarity search, and generating grounded answers via LLaMA 3.1 8B inference on Groq (750+ tokens/sec).
- Achieved sub-50ms vector retrieval latency, significantly outperforming keyword-based (TF-IDF) search on factual accuracy; evaluated with RAGAS framework — Faithfulness 0.91, Answer Relevancy 0.87; deployed as a Streamlit application with real-time token streaming and multi-document ingestion.
- Applied prompt engineering and context-window optimisation (chunk_size=500, overlap=50, k=4) to reduce hallucination rate by 60%+ vs. direct LLM querying, improving answer grounding across research papers, legal documents, and textbooks.

EDUCATION

Bachelor of Technology (B.Tech) – Information Technology

Dr. A.P.J. Abdul Kalam Technical University, Lucknow

CGPA: 8.0

2022 – 2026

ACHIEVEMENTS

- Solved **150+** LeetCode problems (arrays, DP, graphs, trees), reinforcing algorithmic problem-solving for data-intensive engineering challenges — [leetcode.com/u/rajNish_001](#)
- Active **Kaggle** contributor — [kaggle.com/rajneesh08032004](#) — practising end-to-end data science on real-world competition datasets.
- Built and deployed **DocMind** — a production RAG chatbot (LangChain, ChromaDB, LLaMA 3.1/Groq) achieving RAGAS Faithfulness 0.91 — tracking advancements in **Generative AI**, **LLMs**, and **RAG** pipelines.

CERTIFICATIONS

- Machine Learning in Python (**Simplilearn**)
- Machine Learning & Data Science (**GeeksforGeeks**)

Sep 2025
2025 – 2026